
FLiP Data Analysis Pipeline

Custom codes written in Matlab were reported previously (Nature 590, 451-456 (2021)). Using the Matlab files in this folder, along with previous code, follow instructions below.

1 - FLiM Lifetime Fitting (SPCImage)

1.1 Launch SPCImage

Open MATLAB. When the popup appears, click MATLAB.

At the MATLAB prompt, run `spc_drawInit`.

Close figures 1, 2, 4, and 5.

1.2 Load the data and configure the fit

In the `spc_main` window, open the data file:

Go to parent folder/ <basename>.

Select the first full length FLiM file.

Apply the fit settings below, then click Fit with double.

Parameter	Setting
Tau 1	2.14 — checked / <i>fixed for FLiM-AKAR</i>
Tau 2	0.69 — checked / <i>fixed for FLiM-AKAR</i>
Fix — Delta Peak Time	Unchecked for the initial fit (re-check afterward)
Fix — Gauss W	Unchecked for the initial fit (re-check afterward)
Fit start	0.5
Fit end	16
Fit model	Click Fit with double

Table 1. SPCImage fit settings. Tau 1 and Tau 2 are checked as fixed; Delta Peak Time and Gauss W start unchecked for the initial fit.

After, check Fix for Delta Peak Time and Gauss W, and keep them fixed for each subsequent fit.

1.3 Run the per-row analysis

Switch to the MATLAB Command Window.

Set the raw-data folder path: `rawdat_dir = <raw data folder path>`.

Run the row analysis:

`FLiPAnalysisInRow(rawdat_dir, 1, 1, 4)`

#	Argument	Value	Meaning
1	<code>rawdat_dir</code>	path	Raw data folder path
2	<code>binning</code>	1	Seconds of binning
3	<code>channel</code>	1	Acquisition channel
4	<code>start acq</code>	4	First real acquisition number — usually the same as the first large FLiM file number

Table 2. FLiPAnalysisInRow arguments.

Note: The final argument (4) is the first acquisition number you take data from, which is usually the same as the number of the first large FLiM file selected in step 5.

2 - Concatenate FLiP

After lifetime fitting is complete for the dataset, concatenate the per-acquisition FLiP results.

Open MATLAB.

Open the concatenation script, `concatenate_FLiP.m`:

If it is already open, click its `concatenate_FLiP` tab.

Otherwise, navigate to `GitHub > PKA_Sleep > PKA_Sleep > concatenate_FLiP.m`.

Set `rawdat_dir` to the current dataset basename.

Set `acq_range` to the dataset's acquisition range.

Click Save → Run
